ELEG 5633 Detection and Estimation
Minimum Variance Unbiased (MVUB) Estimators

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Outline

- Bayesian Estimation
  - Basic Ingredients of Estimation Theory
  - Minimum Absolute Error (MAE) Estimator
  - Maximum A Posteriori (MAP) Estimator
  - Minimum Mean Square Error (MMSE) Estimator
  - Linear Minimum MSE (LMMSE) Estimators
  - Orthogonality Principle
- Classical Estimation
  - Maximum Likelihood Estimation
  - Minimum Variance Unbiased (MVUB) Estimators

Minimum Variance Unbiased (MVUB) Estimators

- Recall $\mathrm{MSE}(\hat{\boldsymbol{\theta}}) = \|\mathrm{bias}(\hat{\boldsymbol{\theta}})\|_2^2 + \mathrm{var}(\hat{\boldsymbol{\theta}})$
- It is usually impossible to design $\hat{\boldsymbol{\theta}}$ to minimize the MSE because the bias depends on the true value $\boldsymbol{\theta}^*$, which is unknown.
- Restrict to unbiased estimators, $\mathbb{E}(\hat{\boldsymbol{\theta}}) = \boldsymbol{\theta}^*$. Then $\mathrm{MSE}(\hat{\boldsymbol{\theta}}) = \mathrm{var}(\hat{\boldsymbol{\theta}})$
- Note $\mathrm{var}(\hat{\boldsymbol{\theta}})$ does not depend on $\boldsymbol{\theta}^*$.

- A realizable approach: optimize the MSE with respect to all unbiased estimators.
- Minimum Variance UnBiased (MVUB) estimator is defined as

$$\hat{\boldsymbol{\theta}} = \underset{\hat{\boldsymbol{\theta}}:\mathbb{E}(\hat{\boldsymbol{\theta}})=\boldsymbol{\theta}^*}{\mathrm{argmin}} \ \mathbb{E}[\|\hat{\boldsymbol{\theta}} - \mathbb{E}(\hat{\boldsymbol{\theta}})\|_2^2]$$

Example

$X_1, X_2, \ldots, X_n$ i.i.d. $\sim \mathcal{N}(\theta^*, \sigma^2)$. Let $\hat{\theta} = \frac{1}{n}\sum_{i=1}^{n} x_i$. We have

$$\mathbb{E}\hat{\theta} = \theta^*$$

$$\mathsf{MSE}(\hat{\theta}) = \frac{1}{n^2}\sum_{i=1}^{n} \mathsf{var} X_i = \frac{\sigma^2}{n}$$

Is this the MVUB estimator?

This question can be answered by using Cramer-Rao Lower Bound (CRLB).

MVUE

- Does a MVUE always exist?
- If it does, can we always find it?
- Can we say anything about MVUE?

Cramer-Rao Lower Bound (CRLB)

- The CRLB gives a lower bound on the variance of ANY UNBIASED estimator
- Does NOT guarantee the bound can be achieved.
- Can be used to verify that a particular estimator is MVUB.
- Otherwise we can use other tools to construct a better estimator from any unbiased one – Possibly the MVUE if conditions are met.

CRLB for Scalar Parameters

Theorem (Cramer-Rao Lower Bound (CRLB))

Let $p(x|\theta)$ satisfy the regularity condition

$$\mathbb{E}\left[\frac{\partial \ln p(x|\theta)}{\partial \theta}\right] = 0$$

Then the variance of any unbiased estimator $\hat{\theta}$ must satisfy

$$var(\hat{\theta}) \geq \frac{1}{-\mathbb{E}\left[\frac{\partial^2 \ln p(x|\theta)}{\partial \theta^2} \; |_{\theta=\theta^*}\right]} = \frac{1}{\mathbb{E}\left[\left(\frac{\partial \ln p(x|\theta)}{\partial \theta}\right)^2 \; |_{\theta=\theta^*}\right]}$$

Furthermore an unbiased estimator may be found that attains the bound for all θ iff

$$\frac{\partial \ln p(x|\theta)}{\partial \theta} = I(\theta)[g(x) - \theta]$$

for some $g(\cdot)$ and I. That estimator, which is the MVUE, is $\hat{\theta} = g(x)$ and the minimum variance is $1/I(\theta)$.

Example

$X_1, X_2, \ldots, X_n$ i.i.d. $\sim \mathcal{N}(\theta^*, \sigma^2)$. Let $\hat{\theta} = \frac{1}{n}\sum_{i=1}^{n} x_i$. Is it MVUB?

Solution:

$$\log p(\mathbf{x}|\theta) = N \log \sqrt{2\pi\sigma^2} - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(X_i - A)^2$$

$$\frac{\partial}{\partial\theta}\log p(\mathbf{x}|\theta) = \frac{1}{\sigma^2}\sum_{i=1}^{n}(X_i - A)$$

$$\frac{\partial^2}{\partial^2\theta}\log p(\mathbf{x}|\theta) = -\frac{n}{\sigma^2}$$

$$I(\theta^*) = \mathbb{E}\left[\left(\frac{\partial \log p(\mathbf{x}|\theta)}{\partial\theta}\right)^2 |_{\theta=\theta^*}\right]$$

$$= \frac{1}{\sigma^4}\sum_{i=1}^{n}\mathbb{E}[(x_i - \theta^*)^2] = \frac{n}{\sigma^2} = -\mathbb{E}\left[\frac{\partial^2 \ln p(x|\theta)}{\partial\theta^2} \; |_{\theta=\theta^*}\right]$$

$$\text{var}\hat{\theta} \geq \frac{1}{I(\theta^*)} = \frac{\sigma^2}{n}$$

More about $I(\theta)$

- Fisher Information

$$I(\theta) = -\mathbb{E}\left[\frac{\partial^2 \ln p(x|\theta)}{\partial \theta^2}|_{\theta=\theta^*}\right] = \mathbb{E}\left[\left(\frac{\partial \ln p(x|\theta)}{\partial \theta}\right)^2|_{\theta=\theta^*}\right]$$

- It is always non-negative.
- It is additive for independent observations.
- The CRLB for N i.i.d. observations is $1/N$ times that for one observation.

Theorem (Vector Form of the Cramer-Rao Lower Bound (CRLB))

Assume $p(\mathbf{x}|\boldsymbol{\theta})$ *satisfy the regularity condition* $\mathbb{E}\left[\frac{\partial \ln p(\mathbf{x}|\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\right] = \mathbf{0}, \forall \boldsymbol{\theta}$. *Let* $\hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\theta}}(\mathbf{x})$ *be an unbiased estimator of* $\boldsymbol{\theta}^*$. *Then the error covariance satisfies*

$$\mathbb{E}[(\hat{\boldsymbol{\theta}} - \mathbb{E}\hat{\boldsymbol{\theta}})(\hat{\boldsymbol{\theta}} - \mathbb{E}\hat{\boldsymbol{\theta}})^T] - \mathbf{I}^{-1}(\boldsymbol{\theta}^*) \succeq \mathbf{0}$$

where $\succeq \mathbf{0}$ *means the matrix is positive semi-definite.* $\mathbf{I}(\boldsymbol{\theta}^*)$ *is the Fisher-Information matrix with* (i,j)*th element*

$$\mathbf{I}_{ij}(\theta^*) = -\mathbb{E}\left[\frac{\partial^2 \log p(\mathbf{x}|\boldsymbol{\theta})}{\partial \theta_i \partial \theta_j}|_{\boldsymbol{\theta}=\boldsymbol{\theta}^*}\right]$$

Furthermore an unbiased estimator may be found that attains the bound iff

$$\frac{\partial \ln p(\mathbf{x}|\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = \mathbf{I}(\boldsymbol{\theta})[g(\mathbf{x}) - \boldsymbol{\theta}]$$

In that case, $\hat{\boldsymbol{\theta}} = g(\mathbf{x})$ *is the MVUE with covariance matrix* $\mathbf{I}^{-1}(\boldsymbol{\theta})$.

Example

Consider $x[n] = A + w[n], n = 0, 1, \ldots, N$, where $w[n]$ is WGN with variance σ^2. What is the CRLB for the vector parameter $\boldsymbol{\theta} = [A, \sigma^2]^T$? What the MLE for $\boldsymbol{\theta}$? Is MLE a MVUB estimator?

Efficiency

- An unbiased estimator that achieved the CRLB is said to be efficient.
- Efficient estimators are MVUB, but not all MVUB estimators are necessarily efficient.
- An estimator $\hat{\theta}_n$ is said to be asymptotically efficient if it achieves the CRLB, as $n \to \infty$.
- Recall that under mild regularity conditions, the MLE has an asymptotic distribution

$$\hat{\boldsymbol{\theta}}_n \sim \mathcal{N}\left(\boldsymbol{\theta}^*, \frac{1}{n} I^{-1}(\boldsymbol{\theta}^*)\right) \quad asymptotically$$

so $\hat{\boldsymbol{\theta}}_n$ is asymptotically unbiased.

$$\text{var}(\hat{\boldsymbol{\theta}}_n) = \frac{1}{n} I^{-1}(\boldsymbol{\theta}^*)$$

so it is asymptotically efficient.

Finding the MVUE

We wish to estimate the parameter $\boldsymbol{\theta}$ from the observation $\mathbf{x}$.

1. Determine a sufficient statistic $T(\mathbf{x})$ for the parameter to be estimated $\boldsymbol{\theta}$. This may be done using the Neyman-Fisher factorization theorem.
2. Determine whether the sufficient statistic is also complete. If it is, continue to step 3. Otherwise, we can say nothing more about the MVUE.
3. Find the MUVE $\hat{\boldsymbol{\theta}}$ from $T(\mathbf{x})$ using the Rao-Blackwell-Lehmann-Scheffe theorem
   a) Find a function $g(\cdot)$ of the sufficient statistic that yields an unbiased estimator $\hat{\boldsymbol{\theta}} = g(T(\mathbf{x}))$.
   - By definition of completeness of the statistic, the will yield the MVUE.

   b) Find any unbiased estimators $\tilde{\boldsymbol{\theta}}$ for $\boldsymbol{\theta}$, and then determine $\hat{\boldsymbol{\theta}} = \mathbb{E}[\tilde{\boldsymbol{\theta}}|T(\mathbf{x})]$.
   - This is usually very difficult to do. The expectation is taken over the distribution $p(\tilde{\boldsymbol{\theta}}|T(\mathbf{x}))$

Sufficient Statistics

- Let $\mathbf{x} = [x_1, x_2, \ldots, x_n]$ be a vector containing n i.i.d observations generated according to $p(x; \theta)$.
- Question: Is there a lower dimensional function of $\mathbf{x}$, say $T(\mathbf{x})$, that alone carries all the relevant information about θ?
- Goal: Given a family of distributions $\{p(x; \theta)\}_{\theta \in \Theta}$ and one or more observations from a particular distribution $p(x; \theta^*)$ in this family, find a data compression strategy that preserves all information pertaining to θ^*.
- The function identified by such strategy is called a sufficient statistic.

Definition

Let X denote a random variable whose distribution is parametrized by $\theta \in \Theta$. Let $p(x;\theta)$ denote the PDF. A statistic $T(X)$ is sufficient for θ if the distribution of X given $T(X)$ is independent of θ; i.e.,

$$p(x|T(x) = T_0; \theta) = p(x|T(x) = T_0)$$

Example

Consider $x[n] = A + w[n], n = 0, 1, \ldots, N-1$, where $w[n]$ is WGN with variance σ^2. Assume σ^2 is known. What are examples of sufficient statistics for A?

Note that

$$\begin{aligned} p(x|T(x) = T_0; \theta) &= \frac{p(x, T(x) = T_0; \theta)}{p(T(x) = T_0; \theta)} \\ &= \frac{p(x;\theta)\delta(T(x) = T_0)}{p(T(x) = T_0; \theta)} \end{aligned}$$

We can use this formula to verify whether a statistic is sufficient.

Theorem (Neyman-Fisher Factorization Theorem)

Let X be a random variable with density $P(x;\theta)$ for some $\theta \in \Theta$. The statistic $T(X)$ is sufficient for θ iff the density can be factorized into a function $h(x)$ and a function $g(T(x);\theta)$,

$$p(x;\theta) = g(T(x);\theta)h(x)$$

where $g(\cdot)$ is a function depending on x through $T(x)$ and $h(\cdot)$ is a function depending only on x.

Definition

A sufficient statistic is minimal if the dimension of $T(X)$ cannot be further reduced and still be sufficient.

Example

Consider $x[n] = A + w[n], n = 0, 1, \ldots, N-1$, where $w[n]$ is WGN with variance σ^2. Assume σ^2 is known. Find a sufficient statistic for A using N-F factorization.

Example

Consider $x[n] = w[n], n = 0, 1, \ldots, N-1$, where $w[n]$ is WGN with variance σ^2. Find a sufficient statistic for σ^2.

Sufficient Statistics and MVUEs

- If we have determined a sufficient statistic $T(\mathbf{x})$ for θ, we can use this to improve any unbiased estimator of θ, as is proven in the RBLS theorem.
- If we are lucky and the statistic is also complete, we can use it to find the MVUE.
- A statistic is called complete if only one function of it yields an unbiased estimator of θ.
- Another way to check if a statistic is complete is
  A statistic T is complete if $\int_{-\infty}^{\infty} v(T)p(T;\theta)dT = 0$ for all θ is only satisfied by $v(T) = 0$ for all T.

An Example of Incomplete Statistic

Example

Consider estimating A from

$$x = A + w, \quad w \sim U[-1/2, 1/2]$$

Is x a sufficient statistic? Is it complete?

RBLS Theorem

Theorem

If $\tilde{\boldsymbol{\theta}}$ is an unbiased estimator of θ and $T(\mathbf{x})$ is a sufficient statistic for θ, then $\hat{\boldsymbol{\theta}} = \mathbb{E}[\tilde{\boldsymbol{\theta}}|T(\mathbf{x})]$ is

1. *a valid estimator for θ (i.e., it does not depend on $\boldsymbol{\theta}$)*
2. *unbiased*
3. *of lesser or equal variance than that of $\tilde{\boldsymbol{\theta}}$ for all $\boldsymbol{\theta}$.*

Additionally, if the sufficient statistic is complete, then $\hat{\boldsymbol{\theta}}$ is the MVUE of $\boldsymbol{\theta}$.

$$\mathbb{E}(\mathbb{E}(X|Y)) = \mathbb{E}(X)$$
$$\text{var}(X) = \mathbb{E}(\text{var}(X|Y)) + \text{var}(\mathbb{E}(X|Y))$$

Conditioning reduces variance!

If $T(\mathbf{x})$ is complete, there is but one function of T that is an unbiased estimator. Therefore, $\hat{\theta}$ is unique, independent of the $\tilde{\theta}$ we choose.

An alternative approach: since there is only one function of the sufficient statistic that leads to an unbiased estimator, we need only find the unique g to make the sufficient statistic unbiased.

Example

Consider $x[n] = w[n], n = 0, 1, \ldots, N-1$, where $w[n] \sim U[0, \beta]$ for $\beta > 0$. Find the MVU estimator for the mean $\theta = \beta/2$.

Note: We cannot use the CRLB for this problem, since the PDF does NOT satisfy the required regularity condition, i.e., $\mathbb{E}\left[\frac{\partial \ln p(x;\theta)}{\partial \theta}\right] \neq 0$.

Exponential Families

Definition

A family of distributions is said to form an exponential family if the distribution $p(x;\theta)$ (the parameter θ may be multi-dimensional) has density of the form

$$p(x;\theta) = \exp\left(\sum_{i=1}^{s} \eta_i(\theta) T_i(x) - B(\theta)\right) h(x).$$

Write $\eta_i = \eta_i(\theta)$ and write

$$p(x;\eta) = \exp\left(\sum_{i=1}^{s} \eta_i T_i(x) - A(\eta)\right) h(x),$$

where $A(\eta) = B(\theta)$.

Example

Normal distribution $p(x; \mu, \sigma^2) \sim \mathcal{N}(\mu, \sigma^2)$

Poisson distribution

$$p(x; \theta) = \frac{e^{-\theta}\theta^x}{x!} = \exp(x \log \theta - \theta)\frac{1}{x!}$$

Binomial distribution

$$p(x; \theta) = \binom{n}{x}\theta^x(1-\theta)^{n-x} = \exp\left(x \log \frac{\theta}{1-\theta} + n \log(1-\theta)\right)\binom{n}{x}$$

Suppose we have n independent samples $X_1, ..., X_n$ from $p(x;\theta)$ in an exponential family. Individually, the density functions are $p(x_k;\theta)$, so the joint density function is

$$\begin{aligned} p(\mathbf{x};\theta) = \prod_{k=1}^{n} p(x_k;\theta) &= \prod_{k=1}^{n} \exp\left(\sum_{i=1}^{s} \eta_i T_i(x_k) - A(\eta)\right) h(x_k) \\ &= \exp\left(\sum_{i=1}^{s} \eta_i \left(\sum_{k=1}^{n} T_i(x_k)\right) - nA(\eta)\right) h(x_1)\dots h(x_k) \\ &= \exp\left(n\sum_{i=1}^{s} \eta_i \tilde{T}_i(\mathbf{x}) - nA(\eta)\right) h(x_1)\dots h(x_k) \end{aligned}$$

where $\tilde{T}_i(\mathbf{x}) = \frac{1}{n}\sum_{k=1}^{n} T_i(x_k)$

Example

If we have n independent samples from a $\mathcal{N}(\mu, \sigma^2)$ distribution then $\tilde{T}_1(\mathbf{x}) = \bar{\mathbf{x}}$ and $\tilde{T}_2(\mathbf{x}) = \frac{1}{n}\sum_{k=1}^{n} x_k^2$

Theorem

For exponential families in natural form

$$p(x;\eta) = \exp\left(\sum_{i=1}^{s} \eta_i T_i(x) - A(\eta)\right) h(x)$$

The statistics $T = [T_1, T_2, \ldots, T_s]$ is sufficient. T is complete if the natural parameter space $\Pi_0 \subseteq \mathbb{R}^s$ contains an s-dimensional rectangle.

Example

For $X_1, \ldots, X_n \sim Bernoulli(\theta)$, $\bar{X}$ is complete and sufficient.

Example

Consider $x[n] = A + w[n], n = 0, 1, \ldots, N-1$, where $w[n]$ is WGN with variance σ^2. Find the MVUB estimator for vector parameter $\boldsymbol{\theta} = [A, \sigma^2]^T$.

Best Linear Unbiased Estimators

So far, we have learned

- CRLB, may give you the MVUE
- RBLS theorem, may give you the MVUE if you can find sufficient and complete statistics
- All assume we know $p(x;\theta)$; MUVE still may be tough to find

Best Linear Unbiased Estimators

- Simplify the process by constraining the estimators to be linear, i.e.,

$$\hat{\theta} = \sum_{n=0}^{N-1} a_n x[n] = \mathbf{a}^T \mathbf{x}$$

- Only need the first and second moments of $p(x;\theta)$, which is fairly practical.
- Trading optimality for practicality. There is no reason to believe that a linear estimator is efficient, an MVUE, or optimal in any sense.

BLUE assumptions

We need only assume the following

- $x[n] = s[n]\theta + w[n], n = 0, 1, \ldots, N-1$ for $s[n]$ known, i.e.,

$$\mathbf{x} = \mathbf{s}\theta + \mathbf{w}$$

  This ensures that we can find a linear unbiased estimator.
- $\mathbf{C} = \mathbb{E}[(\mathbf{x} - \mathbb{E}[\mathbf{x}])(\mathbf{x} - \mathbb{E}[\mathbf{x}])^T]$ is known.

We wish to find the linear unbiased estimator with the minimum variance.

- $\hat{\theta} = \mathbf{a}^T\mathbf{x}$
- $\mathbb{E}[\hat{\theta}] = \theta \Rightarrow \mathbf{a}^T\mathbf{s} = 1$
- $\text{var}(\hat{\theta}) = \mathbf{a}^T\mathbf{C}\mathbf{a}$

BLUE – Scalar Case

$$\begin{aligned} \min_{\mathbf{a}} \quad & \mathbf{a}^T \mathbf{C} \mathbf{a} \\ \text{s.t.} \quad & \mathbf{a}^T \mathbf{s} = 1 \end{aligned}$$

Solving the optimization problem, we have

$$\mathbf{a}_{OPT} = \frac{\mathbf{C}^{-1}\mathbf{s}}{\mathbf{s}^T \mathbf{C}^{-1}\mathbf{s}}$$

$$\text{var}(\hat{\theta}) = \frac{1}{\mathbf{s}^T \mathbf{C}^{-1}\mathbf{s}}$$

Example

$x[n] = A + w[n], n = 0, 1, \ldots, N-1$, $w[n]$ not Gaussian, but independent, identically distributed with zero mean and variance σ^2.

The sample mean is the BLUE independent of the PDF of the data.
It is the MVUE for Gaussian noise.

Example

$x[n] = A + w[n], n = 0, 1, \ldots, N-1$, $w[n]$ not Gaussian, but independent with zero mean and variance σ_n^2.

BLUE vs. MVUE

When does the BLUE become the MVUE?

$$x[n] = A + w[n], w[n] \sim \mathcal{N}(0, \sigma^2)$$
$$x[n] = w[n], w[n] \sim U[0, \beta]$$
$$x[n] = w[n], w[n] \sim \mathcal{N}(0, \sigma^2)$$

BLUE – Vector Case

- $$\mathbf{x} = \mathbf{H}\boldsymbol{\theta} + \mathbf{w}$$

  This ensures that we can find a linear unbiased estimator.
- $\mathbf{C} = \mathbb{E}[(\mathbf{x} - \mathbb{E}[\mathbf{x}])(\mathbf{x} - \mathbb{E}[\mathbf{x}])^T]$ is known.

We wish to find the linear unbiased estimator with the minimum variance for each $\theta_i \in \boldsymbol{\theta}$.

- $\hat{\theta}_i = \mathbf{a}_i^T \mathbf{x}$ for all i, i.e., $\hat{\boldsymbol{\theta}} = \mathbf{A}\mathbf{x}$
- $\mathbb{E}[\hat{\theta}_i] = \theta_i \Rightarrow \mathbf{A}\mathbf{H} = \mathbf{I}$
- $\text{var}(\hat{\theta}_i) = \mathbf{a}_i^T \mathbf{C} \mathbf{a}_i$

Theorem (Gauss-Markov)

If the data are of the general linear model form

$$\mathbf{x} = \mathbf{H}\boldsymbol{\theta} + \mathbf{w}$$

where $\mathbf{H}$ *is a known* $N \times p$ *matrix,* $\boldsymbol{\theta}$ *is a* $p \times 1$ *vector of parameters to be estimated, and* $\mathbf{w}$ *is a* $N \times 1$ *noise vector with zero mean and covariance* $\mathbf{C}$*, then the BLUE of* $\boldsymbol{\theta}$ *is*

$$\hat{\boldsymbol{\theta}} = (\mathbf{H}^T\mathbf{C}^{-1}\mathbf{H})^{-1}\mathbf{H}^T\mathbf{C}^{-1}\mathbf{x}$$

and the minimum variance of $\hat{\theta}_i$ *is*

$$var(\hat{\theta}_i) = [(\mathbf{H}^T\mathbf{C}^{-1}\mathbf{H})^{-1}]_{ii}$$

In addition, the covariance matrix of $\hat{\boldsymbol{\theta}}$ *is*

$$\mathbf{C}_{\hat{\theta}} = (\mathbf{H}^T\mathbf{C}^{-1}\mathbf{H})^{-1}$$

Example

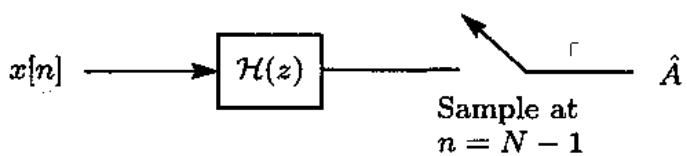
6.11 Consider the problem of estimating a DC level in WSS noise or given

$$x[n] = A + w[n] \qquad n = 0, 1, \ldots, N-1$$

where $w[n]$ is a zero mean WSS random process with ACF $r_{ww}[k]$, estimate A. It is proposed to estimate A by using the output of the FIR filter shown in Figure 6.5 at time $n = N-1$. Note that the estimator is given by

$$\hat{A} = \sum_{k=0}^{N-1} h[k]x[N-1-k].$$

The input $x[n]$ is assumed to be zero for $n < 0$. To obtain a good estimator we wish to have the filter pass the DC signal A but block the noise $w[n]$. Hence, we choose $H(\exp(j0)) = 1$ as a constraint and minimize the noise power at the output at time $n = N-1$ by our choice of the FIR filter coefficients $h[k]$. Find the optimal filter coefficients and the minimum noise power at the output of the optimal filter. Explain your results.



$$\mathcal{H}(z) = \sum_{k=0}^{N-1} h[k]z^{-k}$$

Figure 6.5 FIR filter estimator for Problem 6.11

Example

For the general linear model

$$\mathbf{x} = \mathbf{H}\boldsymbol{\theta} + \mathbf{s} + \mathbf{w}$$

where $\mathbf{s}$ is a known $N \times 1$ vector and $\mathbb{E}[\mathbf{w}] = 0$, $\mathbb{E}[\mathbf{w}\mathbf{w}^T] = \mathbf{C}$. Find the BLUE.

Final is scheduled on May 5, 10:15am to 12:15pm.

- In
  - Bayesian Estimation
    - Maximum A Posteriori (MAP) Estimator: scalar
    - Minimum Mean Square Error (MMSE) Estimator: at most two parameters
    - Linear Minimum MSE (LMMSE) Estimators: at most two parameters
    - Orthogonality Principle
  - Classical Estimation
    - Maximum Likelihood Estimation: scalar; invariance, asymptotics
    - Minimum Variance Unbiased (MVUB) Estimators:
      CRLB: scalar; efficiency
      Sufficient statistics
    - Best Linear Unbiased Estimators (BLUE): scalar
- Out
  - Detection Theory
  - Minimum Absolute Error (MAE) Estimator
  - Complete Statistics
  - RBLS Theorem
- This is a closed book exam. Two 8.5 in. $\times$ 11 in. sheet of notes (both sides) are allowed.